Supplementary Lyapunov analysis of the closed rumor-delay model

Supplement to the TransPINN manuscript

1 Scope and parameter provenance

We quantify infinitesimal perturbation growth in the underlying nine-state delay differential equation (DDE), not in the neural network or an interpolated observation curve. The calculation uses the normalized closed empirical model, $\Lambda = d_h = 0$ and $N = 1$, with the fixed coefficients used in the empirical implementation. Time and exponents are measured in normalized time units and inverse normalized time units, respectively. The horizon of 500 is a mathematical extension of this fixed autonomous model beyond the observation window, not a 500-unit empirical forecast.

The 16 baseline parameter sets comprise the archived TransPINN estimates for all four events and three folds reported in the identifiability analysis, and four independently optimized, training-data-only DDE fits. The latter are included as a sensitivity check, not as uniquely identified physical parameters. Reporting gains do not enter the state dynamics. Full parameter values and hashes are supplied in `output/lyapunov/parameter_sets.csv` and `manifest.json`. These calculations do not characterize every admissible parameter set of the open-population model and do not establish the presence or absence of chaos in measured events.

2 Variational equation and history normalization

Write the fixed-parameter system as

$$\dot{X}(t) = f(X(t), X(t-\tau); \theta), \qquad \dot{v}(t) = A(t)v(t) + B(t)v(t-\tau),$$

where $A = \partial_1 f$ and $B = \partial_2 f$ are evaluated along the numerical DDE trajectory. The largest-exponent estimator propagates one generic separation function and repeatedly normalizes its complete delay history, following the history-space approach discussed in [1]. For $v_t(s) = v(t+s)$, we use

$$\|v_t\|_\tau = \left[\frac{1}{\tau}\int_{-\tau}^{0} \|v(t+s)\|_2^2 \, ds\right]^{1/2}, \qquad \widehat{\lambda}_{b,T} = \frac{\sum_{b<t_k\le T} \log g_k}{t_K - t_{k_0}},$$

where g_k is the norm immediately before renormalization and the denominator is the actual retained integration duration. All stored perturbations in $[t_k - \tau, t_k]$ are divided by g_k. This is a long-time largest-exponent estimator from a generic direction; its finite-time values are not maximum singular-value finite-time exponents. Three independent initial directions check sensitivity to alignment.

State and variational equations are integrated together by an independent fixed-step RK4 method of steps, with linear interpolation of delayed states and perturbations at intermediate stages. The delay spans an integer number of mesh intervals: $h = \tau/\lceil \tau/h_{\max} \rceil$. Linear delayed interpolation limits the overall convergence order; fourth-order accuracy is not claimed. The directional derivative $Av + Bv_\tau$ is evaluated by complex-step differentiation of the analytic right-hand side with imaginary step 10^{-28}. No finite-amplitude subtraction of nearby trajectories is used.

The constant state history is

$$\phi = (0.93, 0.03, 0.015, 0.005, 0.010, 0.005, 0.005, 0.20, 0.20).$$

Every run verifies the strict cohort compatibility inequality

$$\phi_E > p(\phi) I(\phi) \frac{1 - e^{-1.4\tau}}{1.4}.$$

The initial separation function is constant, with a seeded Gaussian direction normalized in history norm. Its seven human components sum to zero. This constraint is maintained explicitly at every step by setting $v_U = -\sum_{j\neq U,\, j\le 7} v_j$, preventing numerical leakage into changes of total population. The media perturbations are unrestricted.

3 Numerical verification and sensitivity design

Each baseline parameter set is integrated to $T = 500$, with the first 100 units discarded for the principal estimate. We compare step caps 0.005, 0.0025 and 0.00125. All 16 sets also use perturbation seeds 11, 22 and 33 at the middle resolution. The nominal renormalization interval is 1, rounded to a mesh multiple. For each event's middle-fold neural parameter set we additionally test intervals 0.5 and 2, two state histories, and delays 0.01, 0.05, 0.10, 0.15, 0.22. In the history tests, E, H, R, D are multiplied by 0.5 or 1.5 and S_u is adjusted to preserve $N = 1$. All other quantities are held fixed. This gives 116 runs, all satisfying history compatibility and numerical positivity checks.

The implementation is checked in three ways:

- The variational directional derivative agrees with a centered finite difference of the independent right-hand side to a maximum absolute error of 1.44×10^{-11} at the test state.
- For $\dot{u} = au + 0.1u(t - 0.073)$, the known dominant exponent is $a + W_0(0.1\tau e^{-a\tau})/\tau$. The stable case $a = -0.3$ gives -0.19854010335; numerical errors for the three step caps are approximately 7.84×10^{-9}, 1.96×10^{-9} and 5.07×10^{-10}. A positive-exponent case $a = 0.1$ is also recovered to within 2×10^{-7}, showing that the method is not constrained to return nonpositive values.
- At the reference parameter vector $(2.5, 1.3, 0.073, 1, 1, 1)$, comparison with an independent adaptive method-of-steps solver over one normalized time unit gives maximum state errors 3.81×10^{-9}, 9.48×10^{-10} and 2.46×10^{-10} as the mesh is refined.

These benchmarks validate the implementation locally; they are not certified error bounds for every nonlinear run. Complete numerical checks are supplied in `numerical_checks.json`.

4 Results and interpretation

Across all 116 runs, $|\widehat{\lambda}_{100,500}| < 2.95 \times 10^{-8}$. The largest mesh-dependent spread for a baseline parameter set is 1.03×10^{-10}. On the later interval $[300, 500]$, the recorded exponents are zero to floating-point resolution. Small positive finite-duration estimates therefore do not persist as the transient is removed. The maximum final $E+H+R+D$ is 1.26×10^{-45}, and the largest final right-hand-side magnitude is 2.50×10^{-14}. Maximum population conservation error is 2.26×10^{-13}; no negative states were found.

The zero limit is consistent with the equilibrium family

$$X^*(s) = (5s/13,\ 8s/13,\ 0,\ 0,\ 0,\ 0,\ 1 - s,\ 0.2,\ 0.2), \qquad 0 \le s \le 1.$$

Differentiating this family with respect to s gives a nonzero constant separation function in the fixed-population tangent space with zero growth. Thus a maximum exponent near zero is expected for a trajectory approaching a stable member of this family; it is not evidence for a chaotic attractor. The structural relation

$$\dot{U} = 0.2E + 0.25R + 0.25D + \Gamma_U H \ge 0$$

also provides a consistency check on the closed-model trajectories. We do not use this observation alone as a global no-chaos theorem.

Conclusion. Under the tested fixed parameters and compatible histories, the numerical trajectories approach the no-spreading equilibrium family and show no sustained positive largest Lyapunov exponent. This is conditional numerical evidence against sustained chaos in these simulations, not a proof for all parameter regimes, an exclusion of finite-time amplification, or an empirical claim that real rumor propagation cannot be chaotic.

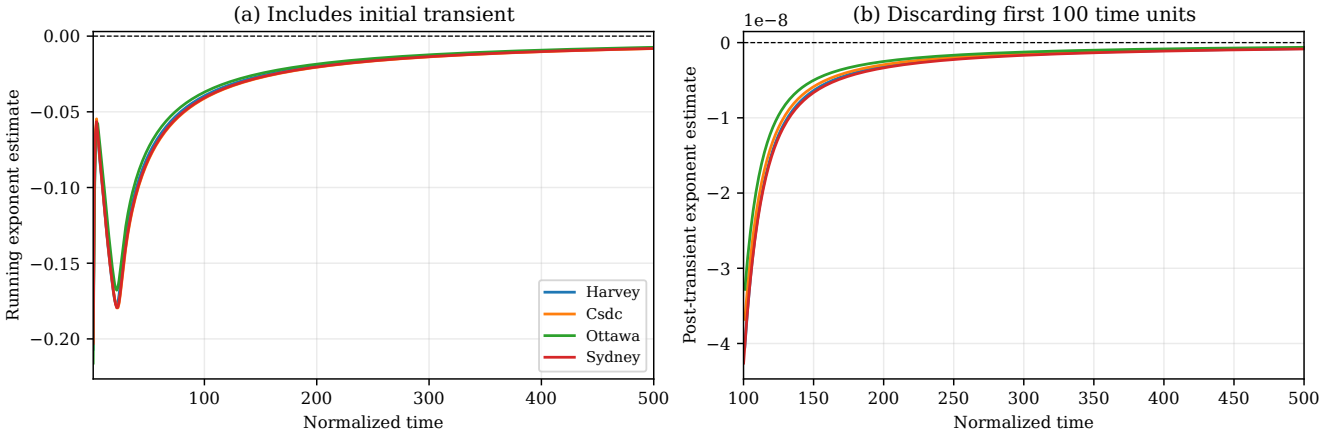


Figure 1: Running generic-direction exponent estimates for the four middle-fold neural parameter sets at the finest mesh. Removing the initial transient makes the approach to the neutral limit explicit. Curves describe the numerical DDE, not network predictions.

5 Baseline convergence and parameters

Parameter set	$\widehat{\lambda}_{100,200}$	$\widehat{\lambda}_{100,300}$	$\widehat{\lambda}_{100,500}$	Mesh spread
harvey_fold1	-3.49e-09	-1.74e-09	-8.72e-10	2.32e-15
harvey_fold2	-3.19e-09	-1.60e-09	-7.98e-10	9.41e-11
harvey_fold3	-3.00e-09	-1.50e-09	-7.50e-10	1.02e-10
csdc_fold1	-3.49e-09	-1.74e-09	-8.72e-10	9.86e-11
csdc_fold2	-2.94e-09	-1.47e-09	-7.34e-10	7.86e-11
csdc_fold3	-4.06e-09	-2.03e-09	-1.02e-09	2.95e-15
ottawa_fold1	-2.28e-09	-1.14e-09	-5.69e-10	6.38e-11
ottawa_fold2	-2.51e-09	-1.25e-09	-6.27e-10	8.15e-11
ottawa_fold3	-3.31e-09	-1.66e-09	-8.28e-10	9.38e-11
sydney_fold1	-3.68e-09	-1.84e-09	-9.21e-10	9.50e-11
sydney_fold2	-3.38e-09	-1.69e-09	-8.46e-10	8.15e-11
sydney_fold3	-3.33e-09	-1.66e-09	-8.31e-10	4.87e-12
harvey_dde_fit	6.87e-08	3.44e-08	1.72e-08	8.54e-16
csdc_dde_fit	-1.11e-08	-5.56e-09	-2.78e-09	9.85e-16
ottawa_dde_fit	-4.81e-11	-2.41e-11	-1.20e-11	3.11e-17
sydney_dde_fit	-1.43e-09	-7.13e-10	-3.56e-10	5.69e-17

The estimates above use direction seed 11 and the finest mesh. Mesh spread is the maximum minus minimum $\widehat{\lambda}_{100,500}$ over the three resolutions. Reported digits facilitate reproducibility and do not imply empirical parameter precision.

Parameter set	β_1	β_2	τ_s	q_R	q_D	χ
harvey_fold1	2.139585	1.095441	0.066639	0.989988	0.993178	1.053143
harvey_fold2	2.259414	1.168851	0.068379	0.994849	0.988663	1.038436
harvey_fold3	2.305371	1.211386	0.073565	0.990600	0.997864	1.046144
csdc_fold1	2.137408	1.106991	0.074204	1.021320	1.066705	1.079482
csdc_fold2	2.346453	1.222994	0.072621	1.004124	1.019770	1.084411
csdc_fold3	2.017474	1.030285	0.074641	1.000947	1.111743	1.089407
ottawa_fold1	2.061437	1.060281	0.069120	1.172469	1.170638	1.201248
ottawa_fold2	2.053829	1.053085	0.069527	1.141909	1.155076	1.188194
ottawa_fold3	2.387527	1.241457	0.072597	1.011047	1.017122	1.069719
sydney_fold1	2.034617	1.053840	0.072697	1.014406	1.102793	1.111592
sydney_fold2	2.331433	1.210274	0.071649	1.002645	1.007501	1.075170
sydney_fold3	2.418403	1.260776	0.073240	1.005572	1.006482	1.058615
harvey_dde_fit	2.881854	5.000000	0.010000	0.050000	1.636241	2.999987
csdc_dde_fit	2.517173	0.050180	0.220000	0.647617	4.000000	3.000000
ottawa_dde_fit	0.100000	0.050000	0.010000	0.050000	2.680029	0.367112
sydney_dde_fit	0.100000	0.050000	0.010000	0.050000	0.050000	0.050000

Reproduction. Run the following commands in order:

```
python experiments/run_lyapunov_analysis.py
python experiments/publish_lyapunov_analysis.py
```

The implementation is in `experiments/lyapunov_dde.py`; per-normalization growth factors, parameter values, convergence summaries and source hashes are archived under `output/lyapunov/`. The numerical integration uses NumPy, SciPy and Numba; JiTCDDE is cited for the history-space method but is not called by this implementation.

References

[1] G. Ansmann, "Efficiently and easily integrating differential equations with JiTCODE, JiTCDDE, and JiTCSDE," *Chaos*, vol. 28, 043116, 2018. doi: 10.1063/1.5019320.